

Cycle-Consistent Adversarial Network (CycleGAN)

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Motivation for CycleGAN

Limitation for Conditional GAN

✓ Paired dataset required

Input Image



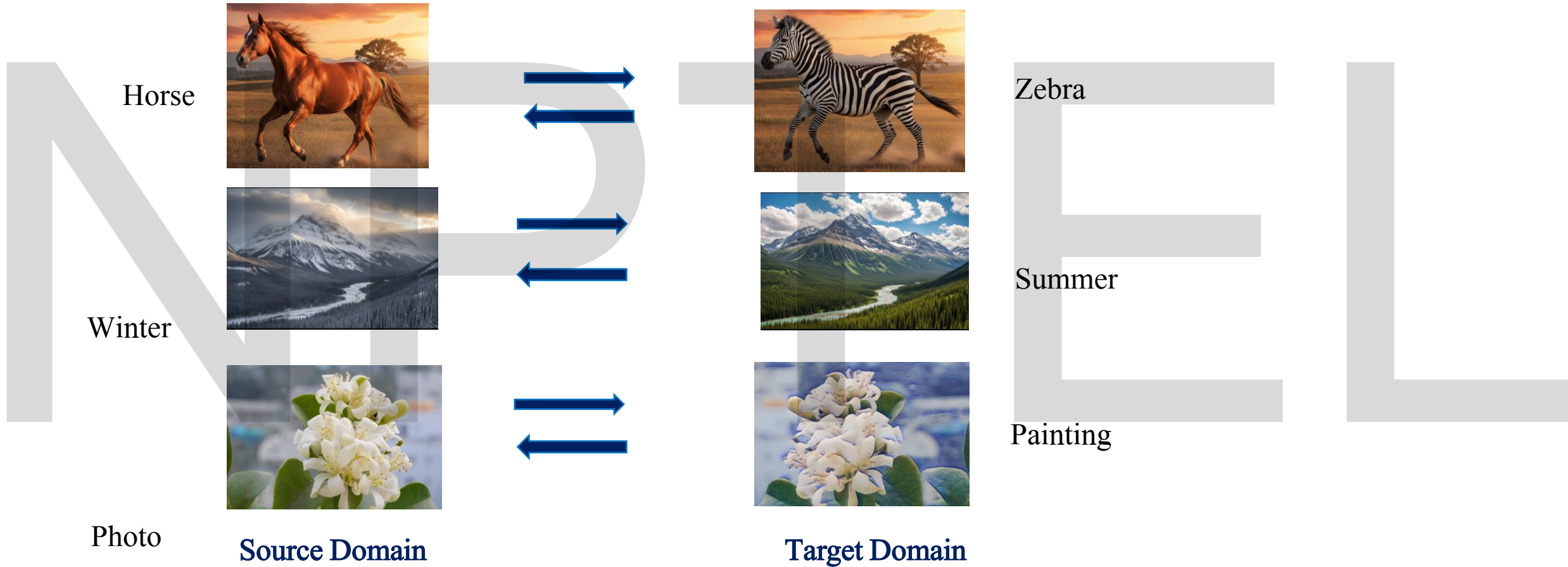
Target Image



- ✓ Pix2Pix requires paired datasets
- ✓ But collecting such paired datasets is difficult.
- ✓ Can a model learn translation between two domains using only separate image collections, without paired examples?

CycleGAN

CycleGAN learns image-to-image translation between two domains without requiring paired examples.



Ref:

Jun-Yan Zhu, Taesung Park, Phillip Isola Alexei A. Efros “Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks”, AI Research (BAIR) laboratory, UC Berkeley

Components of CycleGAN Architecture



Domain X



Domain Y

Two Generators

Generator G: $X \rightarrow Y$
Generator F: $Y \rightarrow X$

Two Discriminators

D_Y : Checks whether zebra images are real or generated.
 D_X : Checks whether horse images are real or generated.

Cycle Consistency
Mechanism

Key idea introduced in CycleGAN

Architecture of CycleGAN-Generators

Learn Two Generators
G and F

- G transforms images from domain X(Horse) to domain Y(Zebra)
- F transforms images from domain Y back to domain X.

Horse



X

Zebra

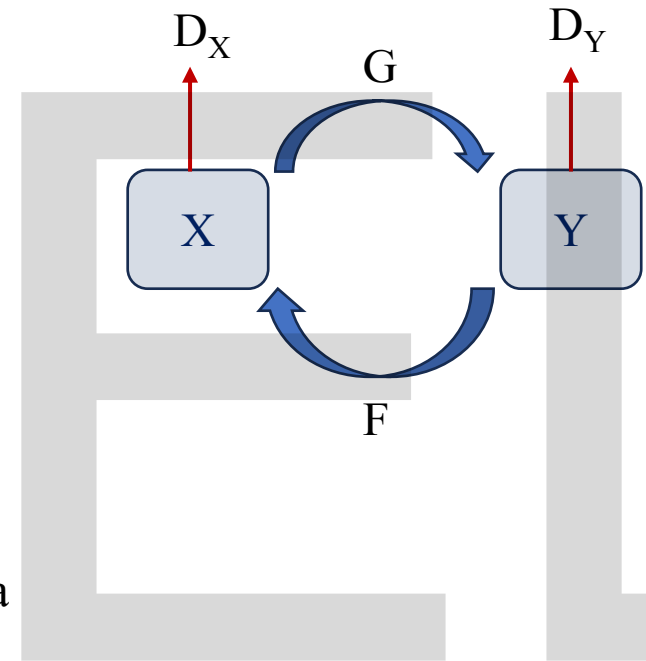


Y

X: Horse
Y: Zebra
G: Horse $\rightarrow$ Zebra
F: Zebra $\rightarrow$ Horse

The generator mapping functions are as follows:

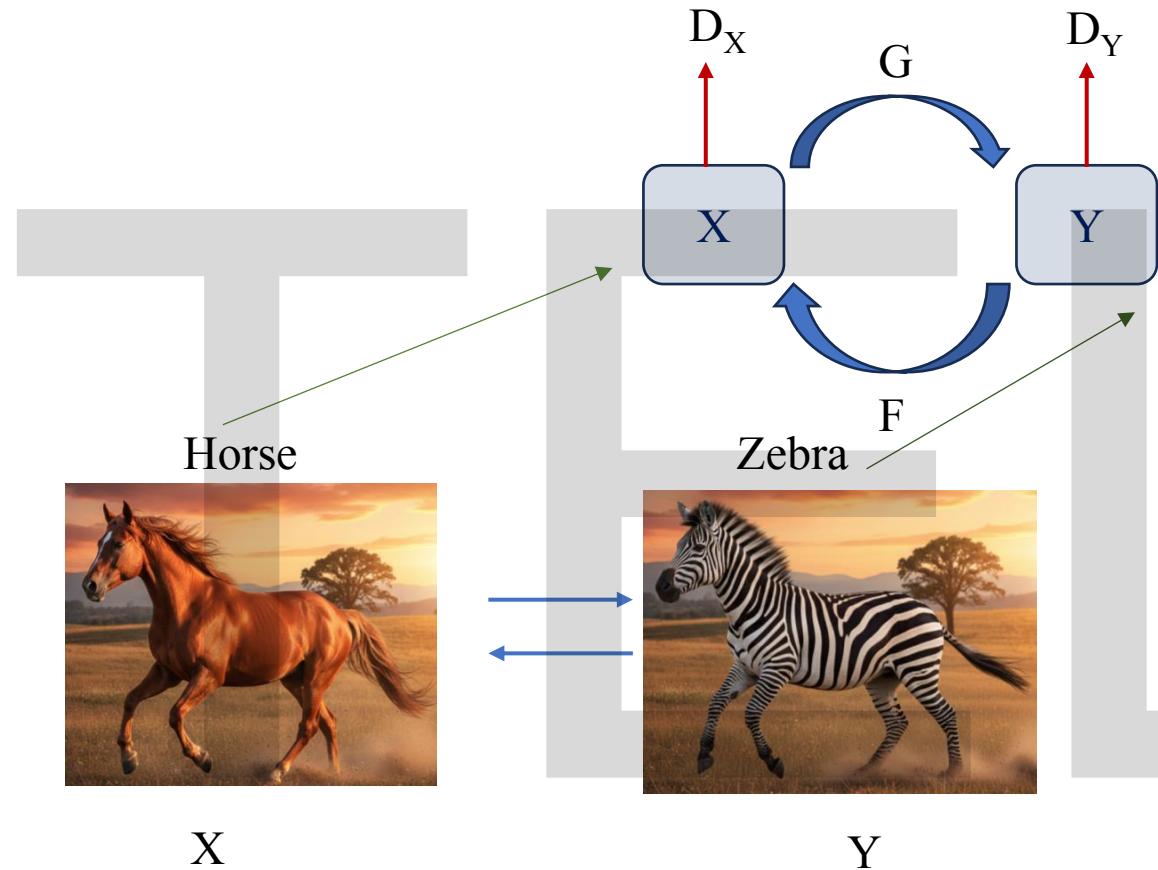
$$G: X \rightarrow Y$$
$$F: Y \rightarrow X$$



Architecture of Cycle GAN- Discriminators

Learn two Adversarial Discriminators
 D_X and D_Y

D_X : Checks whether the horse image is real or fake.
 D_Y : Checks whether the zebra image is real or fake



D_Y :
Distinguishes real images from domain Y and generated images $G(X)$.
 D_X :
Distinguishes real images from domain X and generated images $F(Y)$.

Cycle Consistency in CycleGAN



G
Horse $\rightarrow$ Zebra



F
Zebra $\rightarrow$ Horse



This horse should be same as the original horse



F
Zebra $\rightarrow$ Horse



G
Horse $\rightarrow$ Zebra



This zebra should be same as the original zebra

The translated image should preserve the underlying content/structure of the original image.

Loss Function in CycleGAN

Four Networks are present in Cycle GAN :

G: Translates horse to zebra
 D_Y : Checks if the image of the zebra real or not

F: Translates zebra to horse
 D_X : Checks if the image of the horse real or not

Adversarial Loss

As there are two generators and two discriminators, there are two adversarial losses.

G, D_Y

F, D_X

G: Horse $\rightarrow$ Zebra
F: Zebra $\rightarrow$ Horse

D_X : Is the image of the horse real?
 D_Y : Is the image of the zebra real?

Cycle Consistency Loss

Adversarial Loss

Adversarial Loss from G, D_Y

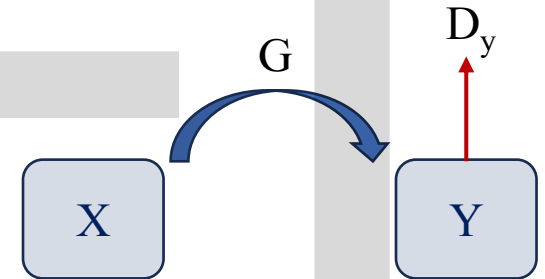
$$L_{GAN}(G, D_Y, X, Y) = \mathbb{E}_{y \sim p_{data}(y)} [\log(D_Y(y))] + \mathbb{E}_{x \sim p_{data}(x)} \log[(1 - D_Y(G(x)))]$$

Trains discriminator D_Y to classify real zebra images as real.

Trains discriminator D_Y to classify generated zebra images as fake.

G: Horse(X) $\rightarrow$ Zebra(Y)

D_Y : whether the zebra image is real or fake?



Desired Output:

$D_Y(\text{fake zebra}) \rightarrow 0$
Generator G tries to fool D_Y
G wants:
 $D_Y(G(x)) \rightarrow 1$

Generator G tries to minimize the loss
Discriminator D_Y tries to maximize the loss

Adversarial Loss

Adversarial Loss from F, D_X

$$L_{GAN}(F, D_X, X, Y) = \mathbb{E}_{x \sim p_{data(x)}} [\log(D_X(x))] + \mathbb{E}_{y \sim p_{data(y)}} [\log(1 - D_X(F(y)))]$$

Trains discriminator D_X to classify real horse images as real.

Trains discriminator D_X to classify generated horse images as fake.

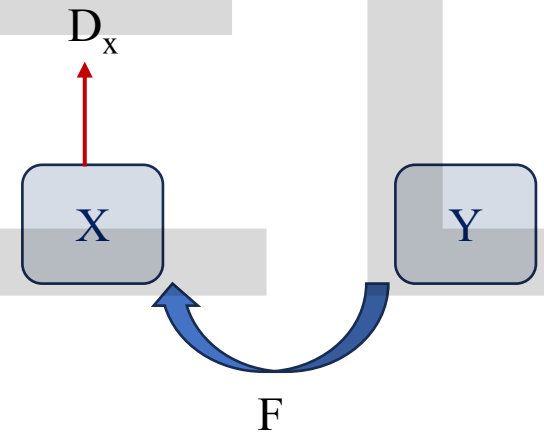
Desired Output:

$D_X(\text{fake horse}) \rightarrow 0$
Generator F tries to fool D_X
 F wants:
 $D_X(F(y)) \rightarrow 1$

- ✓ Adversarial loss only ensures “Output looks realistic”
- ✓ It does NOT ensure Output corresponds to input (any horse could become any zebra)

$F: \text{Zebra}(Y) \rightarrow \text{Horse}(X)$

D_X : Is the image of the horse real?



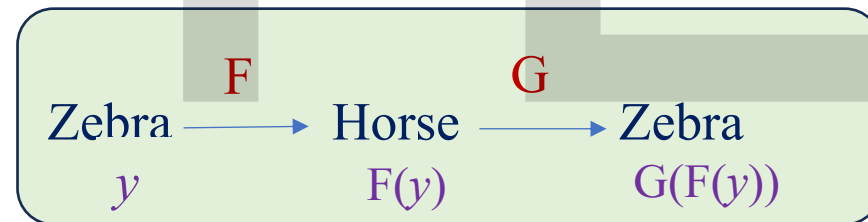
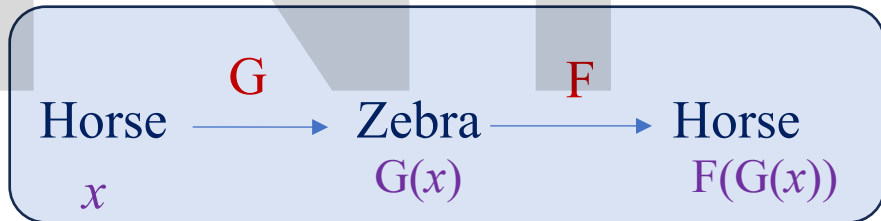
Cycle Consistency Losses in CycleGAN

$$L_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_{\text{data}}(y)} [\|G(F(y)) - y\|_1]$$

Forward Cycle Loss
Original horse vs reconstructed horse

Backward Cycle Loss
Original zebra vs reconstructed zebra

These two errors should be minimized



Objective Function of CycleGAN

Complete CycleGAN loss

$$L(G, F, D_X, D_Y) = L_{GAN}(G, D_Y, X, Y) + L_{GAN}(F, D_X, Y, X) + \lambda L_{cycle}(G, F)$$

Adversarial Loss for horse to zebra

Adversarial Loss for zebra to horse

Cycle consistency Loss
In original paper $\lambda=10$

Objective Function

$$G^* F^* = \arg \min_{G, F} \max_{D_X, D_Y} L(G, F, D_X, D_Y)$$

Find the values of **G and F**
that *minimize* the loss

&

Find **D_X** and **D_Y** that *maximize* the loss

Generators (G, F) try to:

- ✓ Fool discriminators
- ✓ Maintain cycle consistency
→ minimize loss.

Discriminators (D_X, D_Y) try to **correctly classify**
real vs fake
→ maximize loss.

Transformations taken from original paper

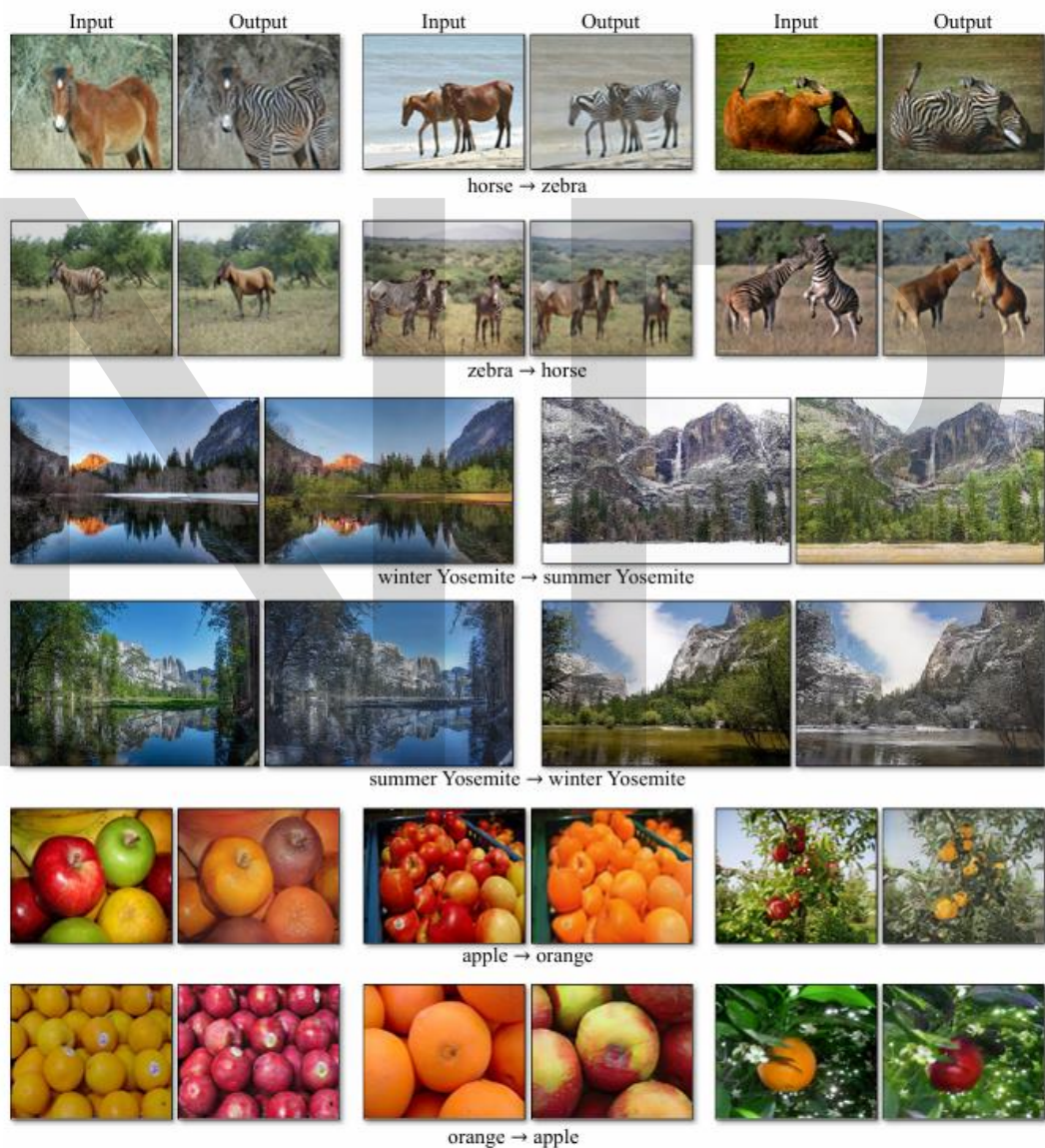


Figure 15: We compare our method with neural style transfer [13] on photo stylization. Left to right: input image, results from Gatys et al. [13] using two different representative artworks as style images, results from Gatys et al. [13] using the entire collection of the artist, and CycleGAN (ours).

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Summary

NIPTEEL

Next Session

StyleGAN

<https://www.youtube.com/watch?v=U8hfliEBcvA>

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